AXEOM STUDIO: SYSTEMS ENGINEERING & FINANCIAL RECOVERY

Executive Operational Audit, Identity Resolution & 6-Pillar Architecture (Pack 175)

PRESENTER INFORMATION & CONTEXT

- **Presenter:** Muhammad Abu Huraira (Computer Science Graduate)
- **Email:** huraira.eqeel@gmail.com
- **Challenge Pack:** Pack 175 | Dallas Creative Studio
- **Business Reference Date:** Tuesday, October 6, 2026 ('Today')
- **Live Web Application:** https://axeom-challenge-five.vercel.app
- **Source Code Repository:** https://github.com/hurairamuzammal/axeom-challenge-pack-175

CORE FINANCIAL RADAR (OCTOBER 6, 2026)

- **Total Invoiced Revenue:** USD 103,890.00 (138 Invoices Across 2026)
- **Bank Deposits Collected:** USD 83,390.00 (122 Reconciled Ledger Payments)
- **Delinquent Overdue Debt:** USD 15,850.00 (8 Invoices Past Net Terms)
- **Trapped Creative Work:** USD 8,210.00 (12 Delivered Cuts Never Billed!)
- **Cash Refund Owed (Blue Bonnet):** USD 0.00 (Completed Rescheduled Shoot B154)
- **Operational Status:** Profitable studio with broken data nervous system

HOW I DIVIDED THE 72-HOUR SPRINT: AN ENGINEERING ROADMAP

Deterministic Systems Engineering vs. Static Spreadsheet Accounting

DAY 1 (HOURS 00 - 24): INGESTION & IDENTITY RESOLUTION

- **Schema Audit:** Audited all 8 raw files line-by-line without modifying originals.
- **Date Normalization:** Diagnosed European DD/MM/YYYY dates in payments.csv vs MM/DD/YYYY.
- **Lead Spam Filter:** Quarantined 4 bot/crypto spam leads (.ru / Telegram) into audit log.
- **Entity Deduplication:** Collapsed 72 raw surface names into 29 canonical client accounts.
- **Agency Payer Isolation:** Separated third-party payer Fieldhouse Marketing from clients.

DAY 2 (HOURS 24 - 48): FINANCIAL & PRODUCTION RECONCILIATION

- **Multi-Pass Ledger:** Reconciled 122 bank payments against 138 invoices with FIFO logic.
- **Revenue Leak Uncovered:** Found USD 8,210.00 in delivered shoots with zero invoices generated.
- **Blue Bonnet Prepayment:** Proved USD 0.00 cash owed; voided duplicate invoice INV-1138.
- **TypeSafe AI Jev:** Integrated Jev (docs.typesafe.ai) for voice note & message triage.
- **The Eli Breakdown:** Audited 106 production cards with 0 notes logged by editor Eli.

DAY 3 (HOURS 48 - 72): REACTIVE WEB APP & MULTI-STAGE TESTING

- **Full-Stack Platform:** Engineered reactive state engine with SQLite persistence & REST API.
- **Responsive UI:** Crafted compact cards with zero wasted void space (250px to 1920px).
- **Sequential Stepper:** Built 6-stage operational drawer (Booked -> Editor -> Intake -> Ledger).
- **Dynamic File Upload:** Engineered live multi-file drag-and-drop instant recalculation.
- **Automated Test Suite:** Authored 31 unit & stress tests passing in 1.0s (100% pass rate).

THE IDEAL AXEOM ARCHITECTURE: 6 PILLARS TO ELIMINATE CHAOS

Permanent Systemic Safeguards to Replace Manual Ad-Hoc Communication

PILLARS 1 & 2: INTAKE GATEKEEPER & RELATIONAL KEYS

- **1. Spam Gatekeeper:** Filters bot scrapers and .ru domains at intake; whitelists boutique domains.
- **2. Relational Primary IDs:** Immutable schema: lead_id -> booking_id -> video_id -> invoice_no -> payment_id.
- **Integrity Safeguard:** Eliminates orphaned bank deposits, unbilled shoots, and untracked studio hours.

PILLARS 3 & 4: REVISION BOTS & DROPDOWN CLIENT MASTER

- **3. WhatsApp/Insta Bots:** Prompts clients for Invoice No or Video ID; Jev transcribes audio voice notes.
- **Direct Editor Routing:** Directly injects revision directives onto assigned editor's board card.
- **4. Single-Source Dropdowns:** Mandates picking clients from validated dropdowns at every pipeline stage.
- **Typo Prevention:** Permanently stops email typos (torresdental.con) and collapses 72 name variants.

PILLARS 5 & 6: BALANCE PRE-CHECK & AUTOMATED DUNNING

- **5. Balance Pre-Check Guard:** Verifies prepaid deposits before billing (stops Blue Bonnet double-billing).
- **6. Smart Payment Reminders:** Triggers polite notices 3d before due date, with 7d/14d/30d aging escalation.
- **Suggest, Never Act Alone:** Requires explicit owner one-click confirmation before any dispatch.

STUDIO FINANCIAL RADAR: DELINQUENT DEBT & UNBILLED REVENUE

Forensic Audit as of October 6, 2026 — USD 24,060.00 Total Recoverable Value

DELINQUENT OVERDUE INVOICES (USD 15,850.00 TOTAL)

- **Apex Fitness (USD 4,500.00):** INV-1099 ($1,500, 46d), INV-1100 ($1,500, 38d), INV-1133 ($1,500).
- **Hillcrest PT (USD 5,000.00):** INV-1116 ($2,500, 21d) & INV-1144 ($2,500) withheld over compliance dispute.
- **Vista Pediatrics (USD 2,700.00):** INV-1121 (8d overdue), billed to agency payer Fieldhouse Marketing.
- **Parkside Chiro (USD 2,700.00):** INV-1126 (1d overdue), billed to agency payer Fieldhouse Marketing.
- **The Ridgeway Lab (USD 900.00):** INV-1134 (Shoot B074, past net terms).

TRAPPED CREATIVE WORK (USD 8,210.00 NEVER BILLED)

- **B020 (Adeyemi Law Firm):** USD 2,500.00 May retainer allocation (Cuts V026 & V029 delivered).
- **B024 (Hillcrest Physical Therapy):** USD 2,500.00 May retainer allocation (Cuts V037-V040 delivered).
- **B080 (The Ridgeway Lab):** USD 900.00 Reels Pack (Cuts V167-V169 delivered July 3).
- **B038, B083, B131 (Crown Church):** USD 810.00 total for 3 podcast shoots (Cuts V082, V175, V288 delivered).
- **B132 (Commercial Project):** USD 1,500.00 Content Day shoot (Cuts V291, V292, V294 delivered).

MONTHLY RETAINER ADVANCE BILLING (1ST OF MONTH)

- **Dr. Adeyemi Law Firm:** Billed 1st of month. Total Paid to Date: USD 9,750.00 (15 shoots).
- **Bayou Wealth Management:** Billed 1st of month. Total Paid to Date: USD 10,000.00 (10 shoots).
- **Hillcrest Physical Therapy:** Billed 1st of month. Total Paid to Date: USD 7,500.00 (15 shoots).
- **Blue Bonnet Realty Proof:** USD 0.00 cash owed (Prepayment P5121 applied to B154; INV-1138 voided).

PRODUCTION TEAM AUDIT: RITA VS. ELI & ZOMBIE TASKS

Why Client Requests Get Lost & The Root Cause of 73-Day Stranded Edits

SENIOR EDITOR RITA (SENIOR STANDARD)

- **Assigned Video Cuts:** 96 Video Cards
- **Revision Notes on Cards:** 100% Documented (0 missing notes)
- **Average Edit Hours:** 4.26 Hours per cut
- **Card Revision Rate:** 15.2% of cuts require revisions
- **Dropped Feedback:** 0 Dropped Notes / 0 Client Complaints
- **Effective Cost per Cut:** USD 191.70 (@ $45/hr rate)

JUNIOR EDITOR ELI (THE 'ELI BLACK HOLE')

- **Assigned Video Cuts:** 106 Video Cards (Joined July 3, 2026)
- **Revision Notes on Cards:** 0 Notes Documented (106 completely blank cards!)
- **Average Edit Hours:** 6.30 Hours per cut (+48% higher labor overhead)
- **Card Revision Rate:** 65.4% of cuts trapped in endless revisions
- **Dropped Feedback:** 6 Audio Voice Notes and DMs completely lost in transit
- **Effective Cost per Cut:** USD 176.40 (@ $28/hr rate, but lower throughput)

THE 2 STRANDED ZOMBIE TASKS RESCUED

- **V121 (River City Auto):** Approved July 17, 2026; left stranded in 'In revision' for 81 days.
- **V152 (The Ridgeway Lab):** Approved July 25, 2026; left stranded in 'In revision' for 73 days.
- **Carmen's Brand Color Note:** Client sent: 'use brand color orange, not the one you used.' Dropped!
- **Operational Resolution:** Jev reconciled approval timestamps; moved cards to Delivered & Billed.

LIVE PRODUCT WALKAROUND: WHAT TO DEMO ON SCREEN

5-Minute Interactive Screencast Map (https://axeom-challenge-five.vercel.app)

1. 6 TOP HERO METRIC TILES

- **Tile 1: Videos (198):** Interactive explorer with instant stage and editor filtering.
- **Tile 2: Clients (24):** Profiles with contact info, total paid, and shoot history.
- **Tile 3: Overdue ($15,850):** Delinquent debt list with aging offsets relative to Oct 6, 2026.
- **Tile 4: Never Billed ($8,210):** Dedicated amber tile clicking directly to unbilled shoots.
- **Tile 5: Pre-Check Guards:** Demonstrates Blue Bonnet Realty $0 cash owed & duplicate voiding.
- **Tile 6: Audit & Decisions:** Visual relational graph and Jev voice note triage queue.

2. RESPONSIVE VIDEO GRID & 6-STAGE DRAWER

- **Compact Responsive Grid:** Tested across 250px mobile, 1280px, 1366px, and 1920px ultrawide.
- **Zero Wasted Void Space:** Shows Video ID, Edit Hours, Revisions, Client, and Status.
- **Amber Action Badges:** Pulsing 'Action Needed: Unbilled ($900)' tags on 12 unbilled cuts.
- **Sequential Stepper Drawer:** Click any card (V152) to trace Booked -> Editor -> Intake -> Approval -> Billing -> Ledger.
- **Retainer Schedule Callouts:** 1st of month advance billing with verified paid-to-date totals.

3. DYNAMIC MULTI-FILE RE-UPLOAD

- **Drag-and-Drop Ingestion:** Click 'Add new files' to upload updated CSVs or messages.txt.
- **Instant Recalculation:** Engine recomputes financial state, debt aging, and queues in 1.1s.
- **Clear Validation Errors:** Surfaces filename, row index, and missing columns on bad files.
- **Suggest, Never Act Alone:** Action cards propose reminder drafts with explicit confirm/reject.

QUESTION 1: WHO ARE OUR REAL CLIENTS? HOW MANY & WHO?

Challenge Brief Part 1 — Finding the Real Clients Across 72 Surface Names

CANONICAL CLIENT PROFILES (29 TRUE CLIENTS)

- **Total Legal Clients:** 29 Canonical Client Accounts (+ 1 Third-Party Agency Payer).
- **Surface Name Fragmentation:** Staff recorded 72 different surface name variants across files.
- **Email Typo Rectification:** Fixed 'torresdental.con' on booking B068 to 'torresdental.com'.
- **Rebrand Continuity:** Linked 'Maple Fitness' historical shoots to 'Maple Athletics'.
- **Personal vs Business Payer:** Mia Torres paying Heart & Hustle Radio invoices via personal Gmail.

THIRD-PARTY AGENCY PAYER ISOLATION (FIELDHOUSE)

- **Payer Identity:** Fieldhouse Marketing (tanya@fieldhousemktg.com) is NOT a client.
- **Agency Role:** Third-party marketing agency settling invoices on behalf of medical clinics.
- **Clinic 1 Covered:** Vista Pediatrics (INV-1121 for USD 2,700.00).
- **Clinic 2 Covered:** Parkside Chiropractic (INV-1126 for USD 2,700.00).
- **Settlement History:** Previously paid 4 invoices in lump-sum wires (P5088 for $5,400).

ANTI-MERGE SAFEGUARDS (SHARED CONTACTS, SEPARATE BUSINESSES)

- **Alana Ibarra Guard:** Operates Hillcrest Physical Therapy (medical) AND Bayou Wealth (finance).
- **Tasha Torres Guard:** Operates Torres Dental PLLC AND Luna Roofing.
- **Nolan Nguyen Guard:** Operates Brightwater (construction) AND Bluebonnet Realty (real estate).
- **Relational Rule:** Isolates tax IDs, domains, and phone numbers to prevent false merges.

QUESTION 2: MONEY. WHO OWES US? DO WE OWE? UNBILLED? ERRORS?

Challenge Brief Part 1 — Reconciled Ledger Audit Totals as of October 6, 2026

WHO OWES US (OVERDUE INVOICES: USD 15,850.00)

- **Apex Fitness (USD 4,500):** INV-1099 ($1,500, 46d), INV-1100 ($1,500, 38d), INV-1133 ($1,500).
- **Hillcrest PT (USD 5,000):** INV-1116 ($2,500, 21d) & INV-1144 ($2,500) withheld over name dispute.
- **Vista Pediatrics (USD 2,700):** INV-1121 (8d overdue), billed to agency payer Fieldhouse.
- **Parkside Chiropractic (USD 2,700):** INV-1126 (1d overdue), billed to agency payer Fieldhouse.
- **The Ridgeway Lab (USD 900):** INV-1134 (Shoot B074, past net terms).

DO WE OWE ANYONE? (USD 0.00 CASH OWED)

- **Blue Bonnet Realty Proof:** Cancelled shoot B142 was rescheduled to B154 and delivered Oct 5.
- **Prepayment Reconciliation:** Wire P5121 ($1,500) applied to completed shoot B154.
- **Duplicate Invoice Voided:** Axeom mistakenly generated INV-1138 ($1,500); voided by our guard.
- **Net Cash Refund Owed:** USD 0.00 cash owed. Zero refund liability.

UNBILLED WORK & BILLING TOOL SYNC ERRORS

- **Trapped Unbilled Work:** USD 8,210.00 across 7 completed shoots (12 delivered cuts): B020, B024, B080, B038, B083, B131, B132.
- **6 Invoices Marked Unpaid:** INV-1066, 1077, 1083, 1084, 1105, 1126 were actually paid in bank deposits.
- **1 Invoice Marked Paid:** INV-1110 (Glow Coffee, $900) marked paid with $0 bank deposit.
- **1 Duplicate Invoice:** INV-1138 ($1,500) issued without checking prepaid deposit.

QUESTION 3: WHY DO CLIENT REQUESTS GET LOST? SINCE WHEN?

Challenge Brief Part 1 — Forensic Evidence & Data Proof from production_board.csv

ROOT CAUSES OF REVISION LOSS

- **1. Disconnected Inbound Channels:** WhatsApp, Instagram DMs, email, and voice notes bypass the board.
- **2. Junior Editor Onboarding Failure:** Editor Eli joined July 3, 2026, and never logged revision notes on cards.
- **3. Vague Phrasing Without IDs:** Clients say 'the one we shot last week' without specifying video IDs.
- **4. Zombie Task Abandonment:** Completed cuts left marked 'In revision' despite client approval.

SINCE WHEN DID IT GET WORSE?

- **Turning Point Month:** July 2026 (specifically July 3, 2026).
- **Trigger Event:** Coincides exactly with junior editor Eli's start date in team.csv.
- **Historical Baseline:** Prior to July, senior editor Rita logged 100% of revision notes.
- **Impact Spike:** Card revision rate jumped from 15% to over 65% under Eli.

DATA PROOF RECEIPTS

- **production_board.csv Audit:** 78.2% of all 202 board cards have empty notes_on_board.
- **Eli's Record:** 0 notes across 106 assigned cards (0.0% coverage); 6.30h average edit time.
- **Rita's Record:** 100% note coverage across 96 assigned cards; 4.26h average edit time.
- **6 Audio Voice Notes:** V049, V135, V132, V268, V274 completely missing from board cards.

QUESTION 4: WHICH CLIENT IS MOST LIKELY TO LEAVE US?

Answered Exclusively via Forensic Receipt Numbers & Audit Trail Proof

TARGET CLIENT & RISK AUDIT

- **Target Account:** Hillcrest Physical Therapy (Dr. Alana Ibarra & Leah Reed)
- **Financial Stake:** USD 5,000.00 withheld cash + USD 30,000.00 annual retainer contract
- **Core Trigger:** Repeated compliance branding errors ignored across 2 months

THE 5 FORENSIC DATA RECEIPTS (CHURN AUDIT TRAIL)

- **Receipt #1 (Audio Voice Note):** 2026-09-02 on B024/V049: 'Please make sure title card says Hillcrest Physical Therapy, not Hillcrest PT. We have strict compliance rules.' Ignored on Eli's card!
- **Receipt #2 (Email Non-Compliance):** 2026-09-08 11:14 CST line 142: 'I noticed new render still says Hillcrest PT. We cannot publish this. Fix immediately.' Ignored!
- **Receipt #3 (Withheld Invoices):** INV-1116 ($2,500, 21d overdue) & INV-1144 ($2,500, unpaid). Client stopped paying retainer!
- **Receipt #4 (Contract Escalation Form):** Lead L097 (2026-09-29 16:45 CST) by Leah Reed asking to discuss contract.
- **Receipt #5 (Urgent Churn Signal):** Email line 189 (2026-10-03 14:22 CST): 'We need to talk about the contract this week.'

ACTION PROTOCOL THIS WEEK (EXECUTION RECEIPTS)

- **Receipt 4a (Executive Conciliation):** Owner Zaid calls Dr. Alana Ibarra & Leah Reed Monday at 9:00 AM to apologize.
- **Receipt 4b (Emergency Re-Render):** Reassign V049 & V135 to senior editor Ali; deliver compliant title cards in 24h.
- **Receipt 4c (Contract & Cash Recovery):** Deliver masters by Tuesday; secure contract renewal before releasing reminder for INV-1116.

QUESTION 5: WHICH MARKETING CHANNEL IS MOST PROFITABLE?

Answered Exclusively via Forensic Receipt Numbers & Audit Trail Proof

MOST PROFITABLE CHANNEL: REFERRAL (91.7% WIN RATE)

- **Receipt #1 (Lead Conversion Audit):** 12 Referral Leads (L003, L012, L019, L027, L034, L045, L058, L069, L074, L086, L091, L095) -> 11 Won Bookings = 91.7% Win Rate!
- **Compared to Paid Channels:** Google Ads: 28.6% win rate | Instagram: 25.0% | Website: 18.2%.
- **Receipt #2 (Revenue & LTV Receipts):** Referrals generated USD 41,400.00 invoiced and USD 33,900.00 cash collected (40.7% of all studio bank deposits!).
- **Retainer LTV Proof:** Dr. Adeyemi (L003) = $9,750 paid; Bayou Wealth (L012) = $10,000 paid.
- **Acquisition Cost (CAC):** USD 0.00 CAC delivering ~100% gross acquisition margin.

WHAT THIS DATA CANNOT TELL US (DATA RECEIPTS)

- **Receipt 5a (Missing Paid Ad Spend):** costs.csv only logs labor rates; it records USD 0.00 ad spend for Google or Meta Ads, making true paid CAC and ROAS uncomputable from files.
- **Receipt 5b (Missing Fixed Overhead):** Facility rent, Dallas utility bills, Sony FX3 camera depreciation, cloud storage, Adobe CC licenses, and insurance are unrecorded.
- **Receipt 5c (Missing Owner Non-Billable Time):** Administrative hours spent by owner Zaid on sales, scheduling, and billing are completely unlogged.
- **Receipt 5d (Missing Formal Contracts & CSAT):** Formal cancellation penalty clauses and NPS customer satisfaction scores do not exist in the data.

QUESTION 6: YOU'RE THE OWNER ON MONDAY MORNING. TOP 3?

Answered Exclusively via Forensic Receipt Numbers & Audit Trail Proof

ACTION 1: CLIENT RESCUE CALL (SAVE $30K ARR & COLLECT $5K)

- **Receipts:** 2026-10-03 email (line 189), Lead L097, withheld retainers INV-1116 ($2,500) and INV-1144 ($2,500), and card errors on V049/V135.
- **Execution:** Owner Zaid calls Dr. Alana Ibarra & Leah Reed at 9:00 AM; assigns 24h compliant re-renders via senior editor Ali.
- **Why First:** Retention precedes acquisition. Saving a $2,500/month retainer protects $30,000/year ARR.

ACTION 2: ISSUE $8,210 UNBILLED INVOICES & VOID DUPLICATE

- **Receipts:** Unbilled shoots B020 ($2,500), B024 ($2,500), B080 ($900), B038 ($270), B083 ($270), B131 ($270), B132 ($1,500); prepayment P5121 linked to completed B154; duplicate INV-1138 ($1,500) voided.
- **Execution:** Dispatch 7 invoices immediately in billing tool; apply credit memo to Blue Bonnet Realty.
- **Why Second:** Immediately captures USD 8,210.00 cash flow for work already completed, with zero incremental labor cost.

ACTION 3: REPLY TO 15 LEADS & CLEAR ZOMBIE TASKS

- **Receipts:** High-intent leads L087 (Mia Torres booking new session), L097 (Leah Reed), L086 (Victor Vega warm referral); zombie board tasks V121 (approved July 17) and V152 (approved July 25).
- **Execution:** Reply to L087, L097, L086 by 11:00 AM; move V121 and V152 to Delivered, unblocking billing.
- **Why Third:** Existing clients and warm referrals convert at over 90% win rate; clearing zombie tasks completes the pipeline.

CAMERA-ON REFLECTION QUESTIONS (SUBMISSION REQUIREMENT)

Spoken Directly into Camera Without Notes — Single Take Delivery

WHAT WAS THE HARDEST PART, AND HOW DID YOU SOLVE IT?

- **The Challenge:** Multi-channel identity resolution without creating false merges across 72 surface names.
- **The Trap:** Three individuals (Alana Ibarra, Tasha Torres, Nolan Nguyen) operate separate legal businesses.
- **The Solution:** Built a 3-tier identity graph separating verified contact anchors, business registries, and third-party agency payers.

WHAT DID YOU GET WRONG AT FIRST?

- **Mistake 1: Blue Bonnet Refund Trap:** Initially thought Axeom owed $1,500 cash for cancelled shoot B142. Chronological audit proved shoot was rescheduled and delivered as B154; true error was duplicate invoice INV-1138.
- **Mistake 2: European Dates:** Standard MM/DD/YYYY parsers scrambled payments.csv dates because it used European DD/MM/YYYY formatting. Enforced dual-format deterministic parsing.

FINAL SYSTEM VERIFICATION & READINESS

- **Automated Test Suite:** 31 unit & scenario stress tests passing in 1.048s.
- **Live URL:** https://axeom-challenge-five.vercel.app
- **Code Repository:** https://github.com/hurairamuzammal/axeom-challenge-pack-175